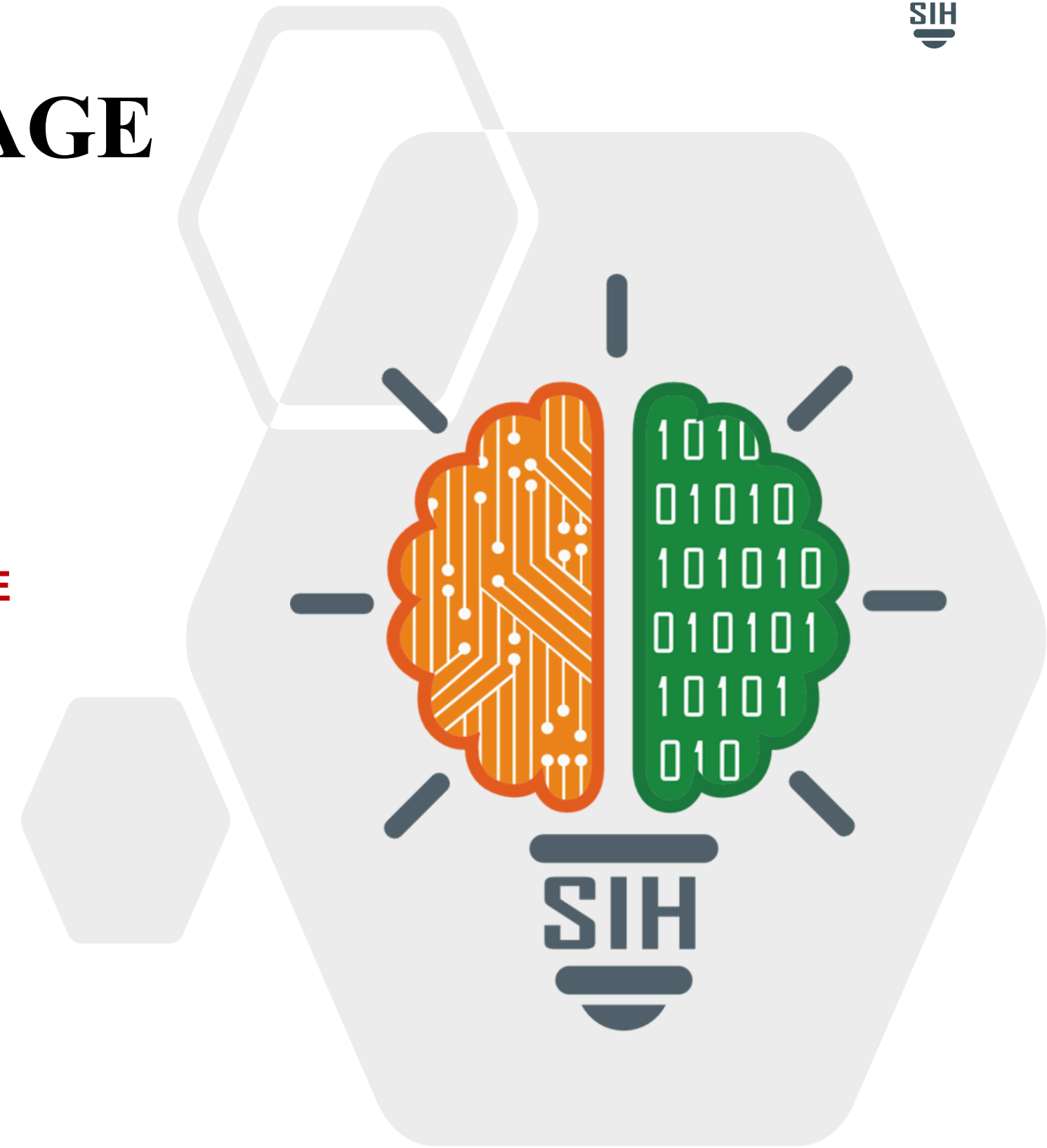


TITLE PAGE

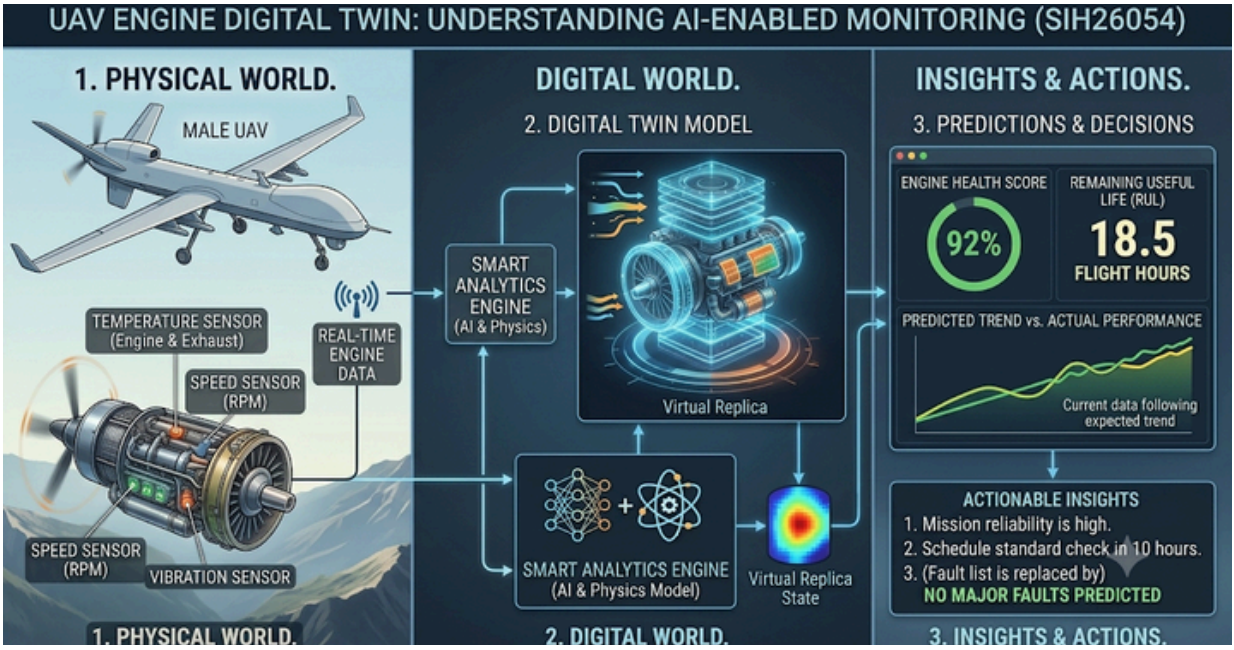
- Problem Statement ID – **SIH26054**
- Problem Statement Title- **AI-Enabled Real-Time Digital Twin System for Health Monitoring, Fault Prediction and Mission Reliability Enhancement of Aero Piston Engines used in MALE UAVs**
- Theme- **Robotics and Drones**
- PS Category- **Software**
- Team ID-
- Team Name :- **CtrlFreaks**





ATLAS is a real-time Digital Twin for UAV aero-piston engines, integrating telemetry, health monitoring, fault detection and predictive analytics.

- **Real-Time Telemetry:** Captures critical engine parameters like RPM, CHT, EGT, oil pressure, fuel flow and vibration for continuous monitoring.
- **Physics + ML Analysis:** Compares actual telemetry with physics-based predictions to detect anomalies, estimate degradation and predict RUL.
- **Backend Integration:** Spring Boot connects the UAV, ML services and database, managing telemetry, engine health and fault information.
- **Unified Dashboard:** Presents real-time and historical data, health status, fault alerts, recommendations and ML predictions in one place



➤ How It Addresses the Problem

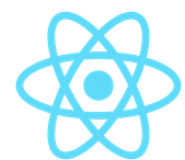
- **Continuous Monitoring:** Tracks critical engine parameters for real-time health monitoring.
- **Early Fault Detection:** Identifies abnormal conditions and generates severity-based alerts.
- **Predictive Maintenance:** Uses ML to detect anomalies, estimate degradation and predict RUL.
- **Physics-Backed Digital Twin:** Compares actual vs. expected engine behaviour to identify deviations early.
- **Mission Reliability:** Uses mission data and telemetry history to analyse engine performance across missions..

➤ Innovation & Uniqueness

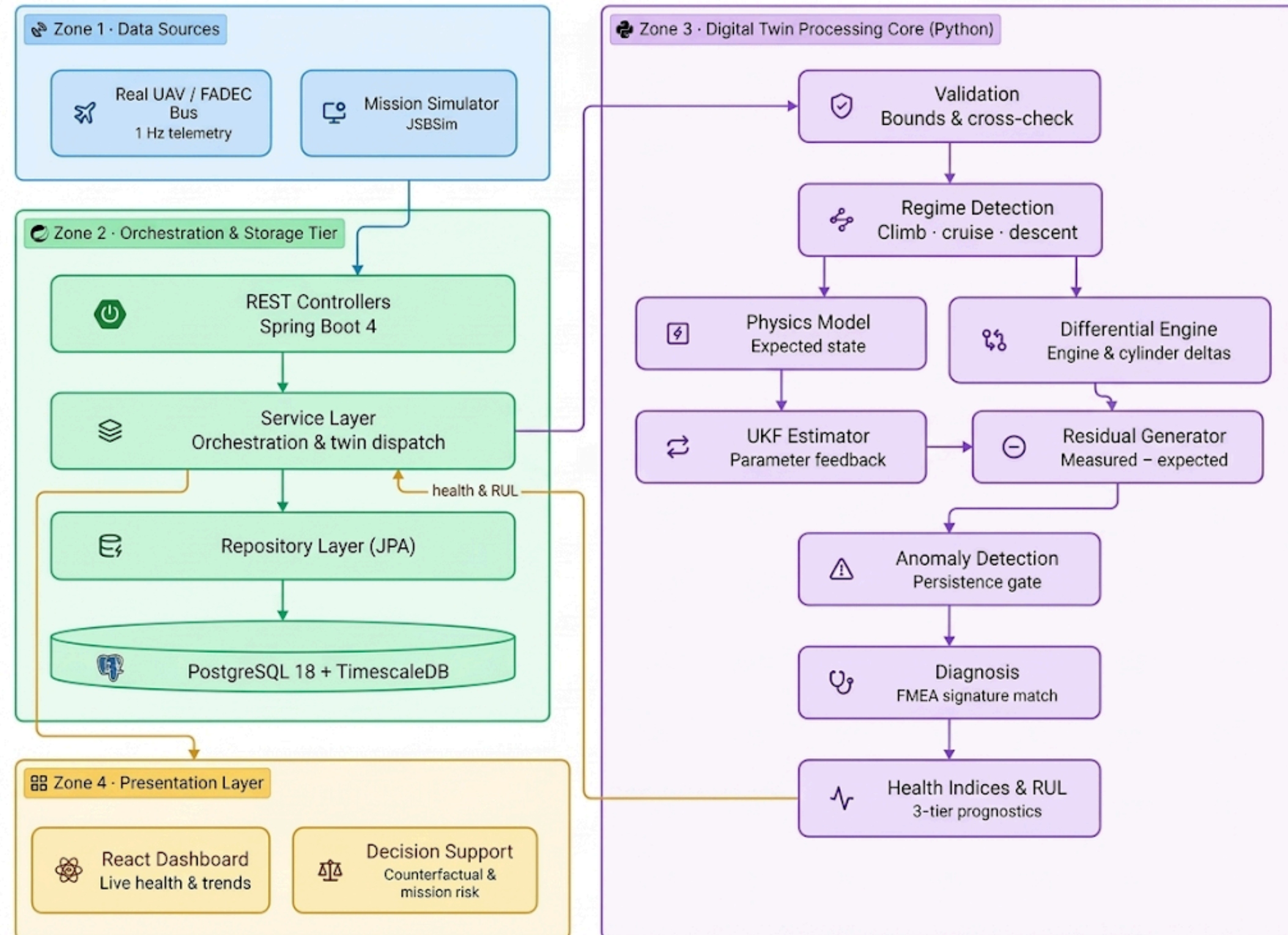
- **Mission-Level Twin:** Analyses engine health across missions and operating conditions using historical telemetry.
- **Actual vs Expected:** Detects early degradation by comparing real and predicted engine behaviour.
- **Fault-Aware Monitoring:** Distinguishes engine faults from sensor anomalies for more reliable predictions.
- **Hybrid Physics + AI:** Combines real telemetry, physics models and machine learning for a smarter Digital Twin.
- **Predictive Intelligence:** Evolves from Telemetry → Fault Detection → Degradation → RUL → Maintenance.



- **Frontend:** React + Vite responsive real-time dashboard UI
- **Visualization:** Recharts live telemetry, equipment health, anomaly & RUL visualizations
- **Backend:** Java + Spring Boot 3.3 REST API, business logic and orchestration
- **Persistence:** Spring Data JPA + Hibernate database access and ORM
- **Twin Integration:** RestTemplate synchronous communication with the Python digital-twin service
- **Digital Twin Service:** FastAPI + Uvicorn high-performance model-serving API
- **Physics & Analytics:** NumPy + SciPy physics-based modeling and degradation calculations
- **State Estimation:** FilterPy + Unscented Kalman Filter (UKF) real-time parameter/state estimation
- **Database:** PostgreSQL relational persistence with JSONB for flexible digital-twin state/configuration
- **Architecture:** Modular microservice architecture with separate UI, Java orchestration layer, Python twin engine, and database.



UAV Engine Digital Twin — Layered Architecture





Technical Feasibility

- **Built on open-source, validated physics (JSBSim)** — no proprietary or classified engine data needed
- **No failure data required by design** — physics-of-failure damage laws + unsupervised anomaly detection sidestep the field's core data-scarcity problem
- **Two independent physics implementations** cross-validate the twin, avoiding circular generate and detect validation
- **Lightweight footprint (~150 bytes/sec/engine)** — runs on existing telemetry links, no new hardware

Implementation

- **Modular three-layer architecture (orchestration dashboard)** — each layer testable and replaceable independently
- **Engine-agnostic configuration** — adapts to different piston-engine profiles without redesign
- **Parameter recovery validated on injected degradation (<4% error)** — proves the estimator works before real data exists

Viability

OPERATIONAL

- Sits above the FADEC, not inside it, no certification burden on flight-critical control, no redesign of existing protection systems
- Fits DRDO's actual constraint, works with zero real flight data, the same constraint every aero-piston PHM system in the field faces
- Engine-agnostic design means one system serves DRDO's varied MALE UAV fleet, not a one-off for a single airframe

ECONOMICAL

- Converts **unplanned engine failure** (asset loss + mission loss) into **scheduled maintenance** (parts + downtime only)
- **No new sensors required** beyond what's already FADEC-instrumented since cost is software, not hardware
- Open-source foundation (JSBSim, Python scientific stack) which means **zero licensing cost** and **no vendor lock-in**

Path to Deployment

- Prototype validates the methodology now, production requires only DRDO supplied bench data and damage constants to calibrate the existing architecture, not a rebuild
- Every assumption in the system is exposed as a labeled, editable parameter meaning DRDO can validate and correct inputs without touching code



IMPACT

- **Longer Mission Confidence:** Helps operators make informed decisions during extended UAV missions.
- **Reduced Downtime:** Identifies developing engine issues early, enabling maintenance before major failures.
- **Lower Lifecycle Cost:** Reduces unnecessary inspections, part replacements and premature engine servicing.
- **Safer Operations:** Provides an additional layer of intelligence between normal operation and critical failure.

BENEFITS

- **For Pilots & Operators:** Clearer visibility into engine condition during missions.
- **For Maintenance Teams:** Early degradation records support targeted inspection and servicing.
- **For Mission Planners:** Engine condition can be considered when making mission and recovery decisions.
- **For Defence Systems:** A scalable digital-health framework that can support future UAV fleets.

USER STORY

SCENARIO

A DRDO MALE UAV is 6 hours into an 18-hour ISR loiter near a border sector. A cylinder-3 injector begins degrading. The onboard FADEC shows nothing every temperature is within its threshold.



IMPACT

ATLAS's twin-engine and cylinder to cylinder differencing flags the divergence over 2 hours before any threshold would trigger, localizing it to cylinder 3 with no fault database and no prior failure data using only the physics of the engine itself.

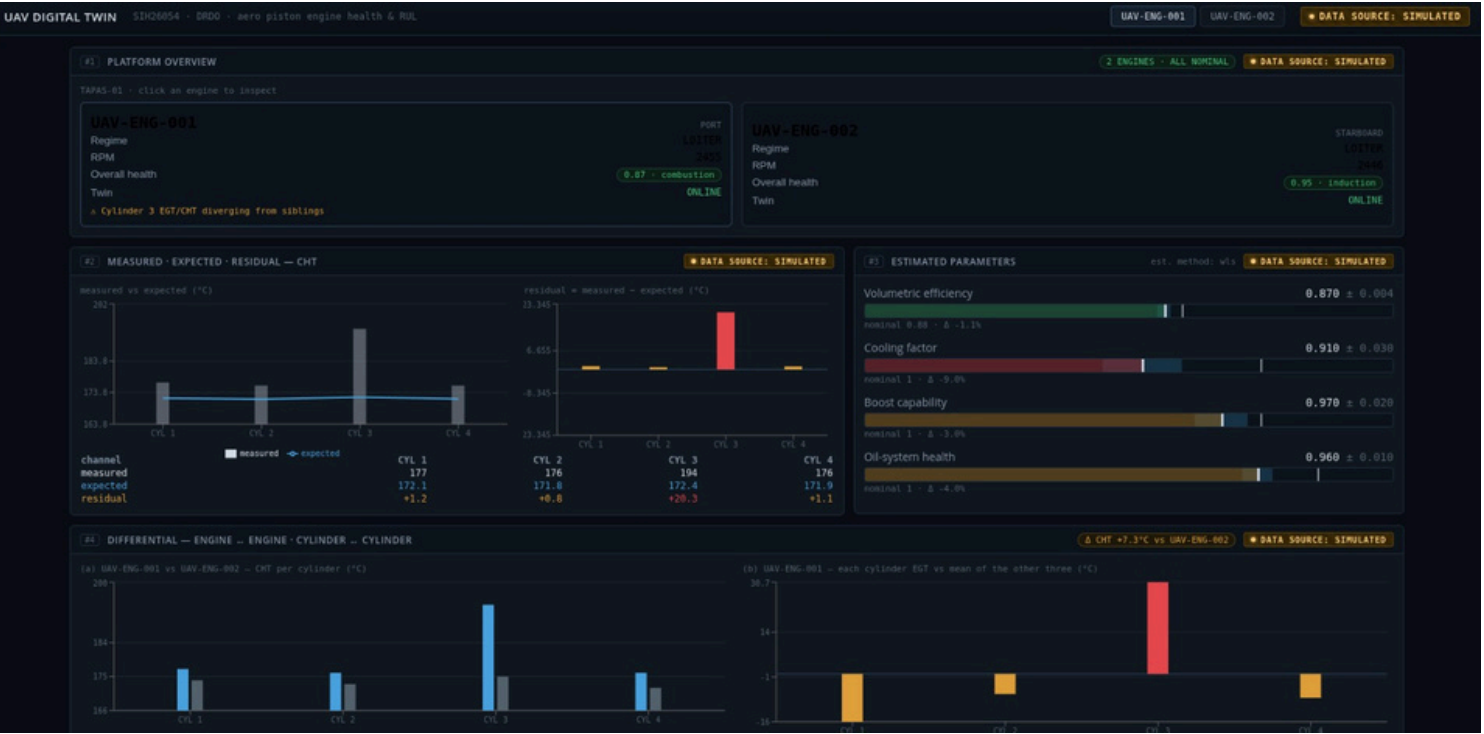


BENEFITS

Ground control receives a mission-aware recommendation with quantified RUL and failure risk, enabling safer decisions while protecting the engine and preserving a maintenance record.

DashBoard UI

RESEARCH AND REFERENCES



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Github repolink

Video link

